

Math Thermo
class # 27
5/05/2026

Isothermal Cahn-Hilliard Revisited

We have derived the Cahn-Hilliard equation previously by starting with two mass fractions ϕ_1, ϕ_2 , eliminating ϕ_1 midway through the calculation.

Here, we eliminate ϕ_1 from the beginning. Suppose

$$\phi_1 + \phi_2 = 1$$

and

$$W(\phi_1, \phi_2) = \frac{\phi_1^2 \phi_2^2}{4}.$$

Set

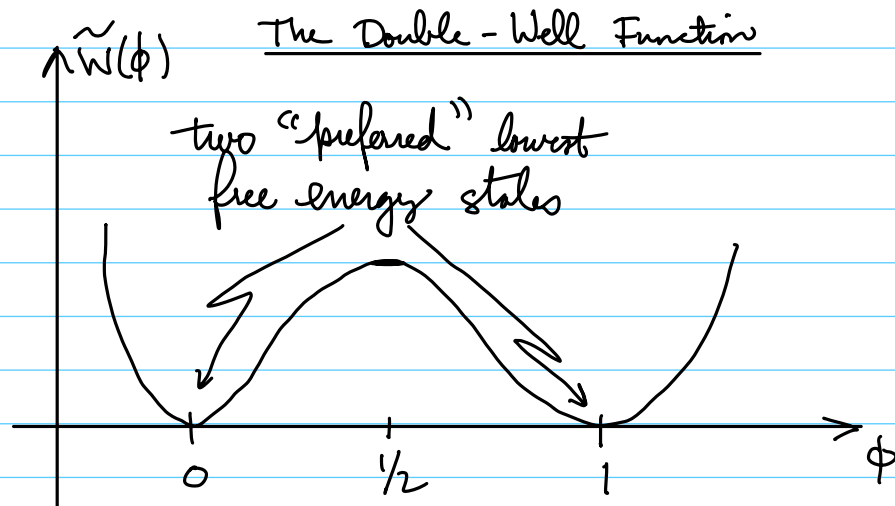
$$\phi := \phi_2$$

and define

$$\tilde{W}(\phi) = W(1-\phi, \phi).$$

Then

$$\tilde{W}(\phi) = \frac{1}{4} (1-\phi)^2 \phi^2$$



Recall

$$\hat{f} = \tilde{f}(T) + \lambda_f \delta(\phi_1, \phi_2, \nabla \phi_1, \nabla \phi_2),$$

where

$$\delta = \frac{1}{\varepsilon} W(\phi_1, \phi_2) + \frac{\varepsilon}{4} (|\nabla \phi_1|^2 + |\nabla \phi_2|^2)$$

and $\varepsilon > 0$ is a parameter with units of length.
Since

$$\phi_1 = 1 - \phi_2 = 1 - \phi$$

$$\begin{aligned} \tilde{\delta}(\phi, \nabla \phi) &= \delta(1 - \phi, \phi, -\nabla \phi, \nabla \phi) \\ &= \frac{1}{\varepsilon} \tilde{W}(\phi) + \frac{\varepsilon}{2} |\nabla \phi|^2. \end{aligned}$$

We will work, for simplicity, under iso-thermal conditions:

$$T \equiv T_0 > 0 \quad (\text{constant}).$$

We also assume that the material is incompressible ($\rho \equiv \rho_0 > 0$, constant), non-reactive, and diffusion dominated. We abuse notation by dropping tildes:

$$\hat{f}(\phi, \nabla \phi) = \tilde{f}_0 + \lambda \delta(\phi, \nabla \phi),$$

where

$$\tilde{f}_0 := \tilde{f}(T_0), \quad \lambda = \lambda_f(T_0) > 0,$$

$$\delta = \frac{1}{\varepsilon} W(\phi) + \frac{\varepsilon}{2} |\nabla \phi|^2,$$

$$W(\phi) = \frac{1}{4} \phi^2 (1 - \phi)^2.$$

Set

$$F = \int_{\Omega} \hat{f}(\phi, \nabla \phi) d\vec{x}.$$

The equation of motion is

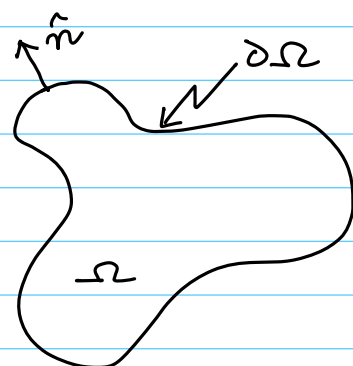
$$\frac{\partial \phi}{\partial t} = -\frac{1}{\rho_0} \nabla \cdot \vec{J},$$

where $\vec{J} = \vec{J}_2^0$. For boundary conditions, let us assume that

$$\hat{n} \cdot \nabla \phi =: \frac{\partial \phi}{\partial n} = 0$$

$$\vec{J} \cdot \hat{n} = 0$$

where $\hat{n}$ is the outward-pointing unit normal on $\partial\Omega$.



To derive the equations, we could find

$$\frac{\partial \hat{f}}{\partial t} = -\nabla \cdot \vec{J}_f^0 + \sigma_f$$

as before. But, I will show you a short-cut. Take the time derivative of F .

$$d_t F = d_t \int_{\Omega} \left\{ \tilde{f}_0 + \lambda \frac{1}{\epsilon} W(\phi) + \lambda \frac{\epsilon}{2} |\nabla \phi|^2 \right\} d\vec{x}$$

$$= \int_{\Omega} \left\{ \frac{\lambda}{\epsilon} W'(\phi) \partial_t \phi + \lambda \epsilon \nabla \phi \cdot \nabla (\partial_t \phi) \right\} d\vec{x}$$

$$= \int_{\Omega} \left\{ \frac{\lambda}{\epsilon} W'(\phi) - \lambda \epsilon \Delta \phi \right\} \partial_t \phi d\vec{x}$$

$$+ \lambda \epsilon \int_{\partial\Omega} \cancel{\frac{\partial \phi}{\partial n}}^{\circ} \partial_t \phi d\vec{x}$$

$$= \int_{\Omega} \lambda W \partial_t \phi d\vec{x}$$

$$= - \int_{\Omega} \frac{\lambda}{\rho_0} W \nabla \cdot \vec{J} d\vec{x}$$

$$= \frac{\lambda}{\rho_0} \int_{\Omega} \nabla \omega \cdot \vec{J} \, d\vec{x} - \frac{\lambda}{\rho_0} \int_{\partial\Omega} \omega \vec{J} \cdot \hat{n} \, d\vec{x}.$$

It is not so easy to identify $\vec{J}_f^D$ with our calculation. However, it is easy to show that

$$d_t F = \int_{\Omega} \sigma_f \, d\vec{x}.$$

Since

$$d_t F = \int_{\Omega} \frac{\lambda}{\rho_0} \nabla \cdot \vec{J} \, d\vec{x},$$

where

$$\sigma_f = \frac{\lambda}{\rho_0} \nabla \omega \cdot \vec{J},$$

$$\omega := \frac{1}{\varepsilon} W'(\phi) - \varepsilon \Delta \phi$$

is the "chemical potential." As usual, we choose

$$\vec{J} = -M \nabla \omega, \quad M > 0 \text{ (constant)}.$$

So that

$$\sigma_f = -\frac{\lambda M}{\rho_0} |\nabla \omega|^2 \leq 0.$$

The equations are

$$\frac{\partial \phi}{\partial t} = \frac{M}{\rho_0} \Delta \omega,$$

(27.1)

in Ω

$$\omega = \frac{1}{\varepsilon} W'(\phi) - \varepsilon \Delta \phi,$$

$$\frac{\partial \phi}{\partial n} = 0 \quad (\text{LTE})$$

on $\partial\Omega$

$$\frac{\partial \omega}{\partial n} = 0 \quad (\text{no flux})$$

The fourth-order, nonlinear, "parabolic" PDE (27.1) is called the binary Cahn-Hilliard equation.

Interfacial Energy

Consider this problem

$$-\varepsilon \frac{d^2 \phi}{dx^2} + \frac{1}{\varepsilon} W(\phi) = 0, \quad -\infty < x < \infty.$$

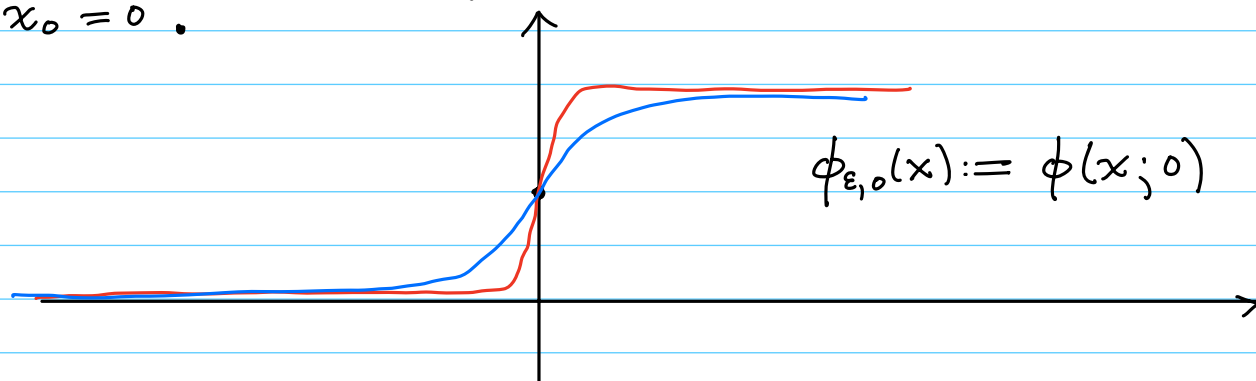
$$W(\phi) = \frac{1}{4} \phi^2 (1-\phi)^2,$$

$$\lim_{x \rightarrow -\infty} \phi(x) = 0, \quad \lim_{x \rightarrow +\infty} \phi(x) = 1.$$

It has a non-unique solution. For any $x_0 \in \mathbb{R}$,

$$\phi(x; x_0) = \frac{1}{2} \left(1 + \tanh \left(\frac{x - x_0}{2\sqrt{2}\varepsilon} \right) \right)$$

is a solution! Try it out! Let us take $x_0 = 0$.



Observe that

$$\int_{-\infty}^{\infty} \frac{\varepsilon}{2} \left(\frac{d\phi_{\varepsilon,0}}{dx} \right)^2 dx = \frac{1}{12\sqrt{2}},$$

$$\int_{-\infty}^{\infty} \frac{1}{\varepsilon} W(\phi_{\varepsilon,0}) dx = \frac{1}{12\sqrt{2}}.$$

In other words, the "chemical part" and the "gradient part" of

$$F[\phi] = \lambda \int_{-\infty}^{\infty} \left\{ \frac{1}{\varepsilon} W(\phi) + \frac{\varepsilon}{2} \left(\frac{\partial \phi}{\partial x} \right)^2 \right\} dx$$

balance. It can be proven that, out of all states from

$$A := \left\{ \psi \in C^2(\mathbb{R}) \mid \lim_{x \rightarrow -\infty} \psi = 0, \lim_{x \rightarrow +\infty} \psi = 1 \right\},$$

$\psi(0) = 1/2?$

$\phi_{\varepsilon,0} \in A$ has the lowest (smallest) free energy. Therefore, in the isothermal setting, we can rightly call it the equilibrium state.

We can identify

$$F[\phi_{\varepsilon,0}] = \frac{\lambda}{6\sqrt{2}}$$

as the one-dimensional interfacial free energy at equilibrium. Notice that $F[\phi_{\varepsilon,0}]$ is independent of ε . This is important.

Also observe that

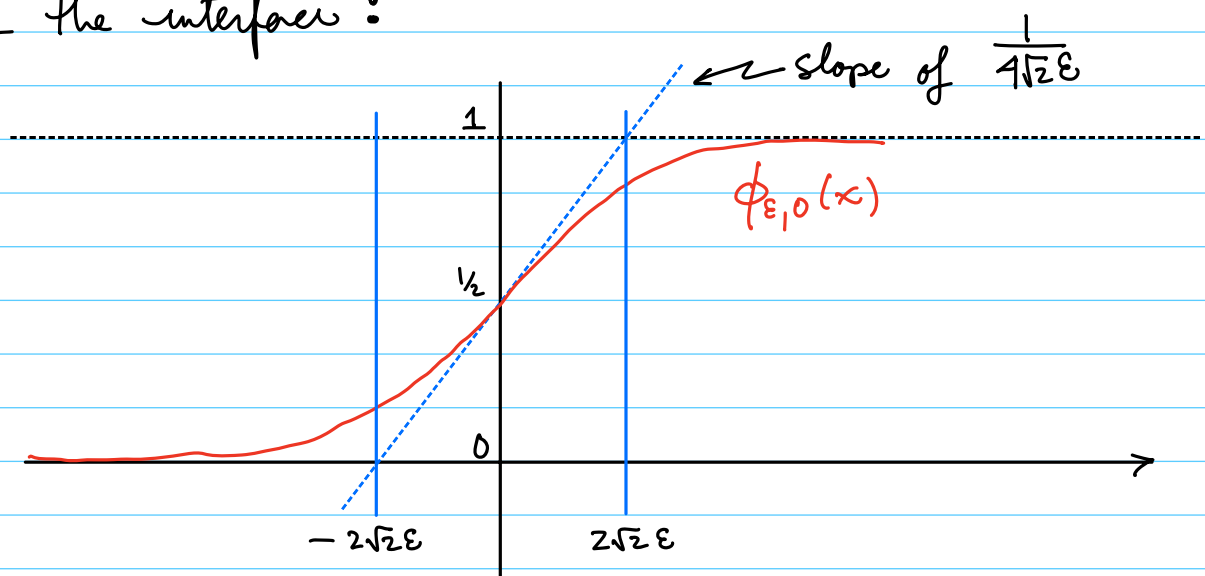
$$\frac{d\phi_{\varepsilon,0}}{dx}(x) = \frac{1}{4\sqrt{2}\varepsilon} \operatorname{sech}^2\left(\frac{x}{2\sqrt{2}\varepsilon}\right).$$

So

$$\frac{d\phi_{\varepsilon,0}}{dx}(0) = \frac{1}{4\sqrt{2}\varepsilon} \xrightarrow{\varepsilon \searrow 0} +\infty.$$

We use this fact to approximate the "thickness"

of the interface :



We say that the diffuse interface thickness is

$$h_\epsilon = 4\sqrt{2}\epsilon,$$

which, as is clear to see, shrinks to zero as $\epsilon \searrow 0$.